OPPOSITE GAPS DEFEAT LARGE-TIME UNIMODALITY FOR ONE-SIDED STABLE PROCESSES

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ABSTRACT. Hasebe and Ueda conjectured that if S_t (resp. T_t) is the law at time t of a classical (resp. free) strictly α-stable process and the initial law μ has a finite absolute moment of order $2+\alpha$, then $\mu * S_t$ (resp. $\mu \boxplus T_t$) is unimodal for all large t. We show that this fails for every $\alpha \in (0,1)$ in both settings, for the stable laws concentrated on the positive half-line. There is one probability measure μ_*, with $\int e^{\lambda|x|}\, d\mu_*(x) < \infty$ for every $\lambda > 0$, such that $\mu_* * S_t^{(\alpha)}$ and $\mu_* \boxplus T_t^{(\alpha)}$ are not unimodal for any $\alpha \in (0,1)$ and any $t > 0$. Its atoms run to $-\infty$, against the direction of the kernel, with gaps tending to infinity. In the classical case the convolution density is strictly decreasing on intervals arbitrarily far to the left; in the free case the density of the free convolution vanishes on an interval lying between two points where it is positive, which we read off from the Huang–Wang description of the support of a free convolution with a freely infinitely divisible law.

1. INTRODUCTION

A probability measure ν on $\mathbb{R}$ is *unimodal* if there are $a \in \mathbb{R}$, $c \geq 0$ and a function $f \geq 0$, nondecreasing on $(-\infty, a)$ and nonincreasing on (a, ∞), with $\nu = c\,\delta_a + f(x)\,dx$. This is the definition used in [5].

Hasebe and Ueda [5] study when a process with independent increments started from an initial law μ becomes unimodal at large times. They show that this happens for free Brownian motion when μ has compact support and for Brownian motion when $\int e^{\beta x^2}\, d\mu(x) < \infty$ for some $\beta > 0$ [5, Theorems 1.2 and 1.3], and they examine the Cauchy process and the positive 1/2-stable process with initial distributions. For the positive 1/2-stable law L_t they prove that $\mu * L_t$ is unimodal for all large t when μ has compact support [5, Theorem 5.3], and they construct μ with $\int |x|^p\, d\mu < \infty$ for all $p < 5/2$ such that $\mu * L_t$ is never unimodal [5, Proposition 5.4]; in that construction the 5/2-moment is infinite. They then write that they conjecture $\mu * L_t$ to be unimodal in large time when the 5/2-moment is finite, and state the general form:

> **Conjecture 5.5 of** [5]**.** Suppose that S_t (resp. T_t) is the law at time $t \geq 0$ of a classical (resp. free) strictly stable process of index $\alpha \in (0,2)$. If μ is a probability measure such that $\int_{\mathbb{R}} |x|^{2+\alpha}\, d\mu(x) < \infty$, then $\mu * S_t$ (resp. $\mu \boxplus T_t$) is unimodal for sufficiently large $t > 0$.

For the symmetric Cauchy law C_t ($\alpha = 1$), for which $\mu * C_t = \mu \boxplus C_t$, a finite third moment does give unimodality at large times [5, Theorem 5.2]. The atoms of the measure in [5, Proposition 5.4] go to $+\infty$, the direction in which the positive kernel spreads mass, and in that construction the positive slope of the density at each new atom has to outweigh the negative slopes of the tails of all the atoms before it; that is what forces the 5/2-moment to be infinite. If the atoms go to $-\infty$ instead, no weight condition is needed at all.

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For $\alpha \in (0,1)$ let $S_t^{(\alpha)}$ be the law with Laplace transform

$$\int_0^\infty e^{-\lambda x}\, S_t^{(\alpha)}(dx) = \exp(-t\lambda^\alpha), \qquad \lambda \geq 0, \tag{1.1}$$

the law at time t of the strictly α-stable subordinator, and let $T_t^{(\alpha)}$ be the law with Voiculescu transform

$$\varphi_{T_t^{(\alpha)}}(z) = -t\, e^{i\pi\alpha} z^{1-\alpha}, \qquad z \in \mathbb{C}^+, \tag{1.2}$$

with the principal branch of the power. This is the positive free strictly α-stable law of Hasebe and Kuznetsov [4], with positivity parameter $\rho = 1$, at time t. Put

$$A_n = 2^{2^n}, \qquad w_n = 2^{-n-2} e^{-A_n^2}\ (n \geq 1), \qquad W = \sum_{n\geq 1} w_n, \qquad \mu_* = (1-W)\,\delta_0 + \sum_{n\geq 1} w_n\, \delta_{-A_n}. \tag{1.3}$$

Theorem 1.1. *The measure μ_* satisfies $\int e^{\lambda|x|}\, d\mu_*(x) < \infty$ for every $\lambda > 0$. For every $\alpha \in (0,1)$ and every $t > 0$, neither $\mu_* * S_t^{(\alpha)}$ nor $\mu_* \boxplus T_t^{(\alpha)}$ is unimodal. In particular Conjecture 5.5 of* [5] *fails for every $\alpha \in (0,1)$, both for classical and for free strictly stable processes.*

At $\alpha = 1/2$, $S_t^{(1/2)} = L_{t/\sqrt{2}}$ in the notation of [5], so the theorem also answers the question raised for L_t just before Conjecture 5.5: a finite 5/2-moment, and even finite exponential moments of every order, do not make $\mu * L_t$ unimodal at large times. Since a compactly supported μ does [5, Theorem 5.3], for this kernel compact support of μ suffices, while no moment or exponential-moment condition on μ does.

The classical half rests on one observation (Lemma 2.1): a kernel that vanishes on $(-\infty, 0]$ and is strictly decreasing on (b, ∞) cannot hide atoms that run to $-\infty$ with growing gaps, because in each gap the density is a sum of decreasing pieces. The free half uses the subordination function of the free convolution and the description of its zero set by Huang and Wang [6]: at the midpoint of each gap the subordination map has a positive real derivative, which puts a whole interval into the zero set of the density, while the atoms on either side of the gap stay in its support.

The theorem leaves open the cases $\alpha \in [1,2)$ and the stable laws whose support is all of $\mathbb{R}$, in particular the symmetric ones; for those the opposite-gap construction does not apply, since the stable law then has a density that is positive on all of $\mathbb{R}$.

2. The classical case

Lemma 2.1. *Let g be a bounded continuous probability density on $\mathbb{R}$ with $g = 0$ on $(-\infty, 0]$, strictly decreasing on (b, ∞) for some $b \geq 0$. Let $0 = a_0 < a_1 < a_2 < \cdots$ with $a_n - a_{n-1} \to \infty$, let $p_n > 0$ with $\sum_{n\geq 0} p_n = 1$, and put $\mu = \sum_{n\geq 0} p_n \delta_{-a_n}$. Then $\mu * g$ is not unimodal.*

Proof. The measure $\mu * g$ has density $q(x) = \sum_{j\geq 0} p_j g(x + a_j)$; the series converges uniformly because g is bounded, so q is continuous, and $\int q = 1$ by Tonelli's theorem. In particular $\mu * g$ has no atom.

Choose N with $a_n - a_{n-1} > b$ for $n \geq N$, and for such n put $I_n = (-a_n + b, -a_{n-1})$, a nonempty open interval. Let $x \in I_n$. For $j \leq n-1$, $x + a_j \leq x + a_{n-1} < 0$, so $g(x + a_j) = 0$ and the term vanishes on I_n. For $j \geq n$, $x + a_j \geq x + a_n > b$, so $x \mapsto g(x + a_j)$ is strictly decreasing on I_n. Hence q is strictly decreasing on I_n (the term $j = n$ has $p_n > 0$).

Suppose $\mu * g = c\,\delta_a + f(x)\,dx$ as in the definition. Since $\mu * g$ has no atom, $c = 0$ and $f = q$ almost everywhere. Let $u < v < a$ and choose sequences $u_k \to u$, $v_k \to v$ in the set where $f = q$, with $u_k < v_k < a$. Then $q(u_k) = f(u_k) \leq f(v_k) = q(v_k)$, and continuity gives $q(u) \leq q(v)$. So q is nondecreasing on $(-\infty, a)$. Since $-a_{n-1} \to -\infty$, some I_n with $n \geq N$ lies in $(-\infty, a)$, where q is strictly decreasing. This is a contradiction. □

Lemma 2.2. *Let $\alpha \in (0,1)$ and $t > 0$. The law $S_t^{(\alpha)}$ has a bounded continuous density g_t with $g_t = 0$ on $(-\infty, 0]$, and there is $b_t \geq 0$ such that g_t is strictly decreasing on (b_t, ∞).*

Proof. By (1.1), $S_t^{(\alpha)}$ is the image of $S_1^{(\alpha)}$ under $x \mapsto t^{1/\alpha}x$, so it suffices to take $t = 1$. A random variable X with law $S_1^{(\alpha)}$ is nonnegative, and its characteristic function is $\exp(-(-iu)^\alpha)$, of modulus $\exp(-\cos(\pi\alpha/2)|u|^\alpha)$, which is integrable since $\cos(\pi\alpha/2) > 0$. Hence X has a bounded continuous density g_1; it vanishes on $(-\infty, 0)$, and by continuity at 0 as well.

For $x > 0$ the density is given by the convergent series [3, Lemma XVII.6.1], [8]

$$g_1(x) = \frac{1}{x}\,E(x^{-\alpha}), \qquad E(y) = \sum_{k\geq 1} c_k y^k, \qquad c_k = \frac{(-1)^{k+1}}{\pi}\,\frac{\Gamma(k\alpha+1)}{k!}\,\sin(k\pi\alpha). \tag{2.1}$$

Since $\Gamma(k\alpha+1)/k! \to 0$ faster than any geometric sequence when $\alpha < 1$, E is an entire function. Differentiating,

$$g_1'(x) = -x^{-2}\big(E(y) + \alpha y E'(y)\big), \qquad y = x^{-\alpha},$$

and $E(y) + \alpha y E'(y) = (1+\alpha)c_1 y + O(y^2)$ as $y \downarrow 0$, with $c_1 = \Gamma(1+\alpha)\sin(\pi\alpha)/\pi > 0$. Hence $g_1'(x) < 0$ for all large x, and g_1 is strictly decreasing on (b_1, ∞) for some b_1; then $b_t = t^{1/\alpha}b_1$ works for $g_t(x) = t^{-1/\alpha}g_1(t^{-1/\alpha}x)$. □

For $\alpha = 1/2$ everything is explicit: $g_t(u) = t\,e^{-t^2/(4u)}/(2\sqrt{\pi}\,u^{3/2})$ for $u > 0$, and

$$g_t'(u) = \frac{t\,e^{-t^2/(4u)}}{2\sqrt{\pi}\,u^{7/2}}\Big(\frac{t^2}{4} - \frac{3u}{2}\Big),$$

so one can take $b_t = t^2/6$. The check script verifies the Laplace transform, this derivative, the scaling, and the series (2.1) against the closed form at $\alpha = 1/2$ and against Kanter's integral representation [7] at $\alpha = 0.3$ and 0.7 (checks K1–K3, S1–S3 in [9]).

Proof of Theorem 1.1, classical part and moments. For every $\lambda > 0$,

$$\int e^{\lambda|x|}\,d\mu_*(x) = 1 - W + \sum_{n\geq 1} 2^{-n-2} e^{\lambda A_n - A_n^2} < \infty,$$

since $\lambda A_n - A_n^2 \leq 0$ once $A_n \geq \lambda$. Also $W \leq \sum_{n\geq 1} 2^{-n-2} = 1/4$, so μ_* is a probability measure with $1 - W \geq 3/4$. Its atoms sit at $-a_n$ with $a_0 = 0$ and $a_n = A_n$, with $a_1 - a_0 = 4$ and $A_n - A_{n-1} = A_{n-1}(A_{n-1} - 1) \to \infty$ for $n \geq 2$. Lemmas 2.1 and 2.2 give that $\mu_* * S_t^{(\alpha)}$ is not unimodal, for every $\alpha \in (0,1)$ and $t > 0$. □

Remark 2.3. The weights in Lemma 2.1 are arbitrary, so no bound on the left tail of μ short of bounded support gives eventual unimodality for these kernels: if $\psi > 0$ is nonincreasing on $[0,\infty)$, take $a_n = A_n$ for $n \geq 1$, $p_n = 2^{-n-1}\min(\psi(A_n), 1)$ for $n \geq 1$ and $p_0 = 1 - \sum_{n\geq 1} p_n$; then $\mu((-\infty, -x)) \leq \sum_{j\geq n} p_j \leq \psi(A_n) \leq \psi(x)$ for $x \in [A_{n-1}, A_n)$ (with $A_0 = 0$), and $\mu * S_t^{(\alpha)}$ is not unimodal for any $\alpha \in (0,1)$ and $t > 0$.

3. The free case

3.1. **Preliminaries.** For a probability measure ν on $\mathbb{R}$ let $G_\nu(z) = \int (z-s)^{-1}\,d\nu(s)$ and $F_\nu = 1/G_\nu$ on $\mathbb{C}^+$, and let φ_ν be its Voiculescu transform, so that $\varphi_{\nu_1 \boxplus \nu_2} = \varphi_{\nu_1} + \varphi_{\nu_2}$ [2]. By the free Lévy–Khintchine theorem [2, Section 5], an analytic function $\varphi : \mathbb{C}^+ \to \mathbb{C}^- \cup \mathbb{R}$ with $\varphi(iy)/(iy) \to 0$ as $y \to \infty$ is the Voiculescu transform of a freely infinitely divisible law. The function in (1.2) qualifies: writing $z = re^{i\theta}$ with $0 < \theta < \pi$, the argument of $-\varphi(z)$ is $\pi\alpha + (1-\alpha)\theta \in (0,\pi)$, and $|\varphi(iy)|/y = ty^{-\alpha} \to 0$. Since $c\,\varphi_{T_1^{(\alpha)}}(z/c) = c^\alpha \varphi_{T_1^{(\alpha)}}(z)$, the law $T_t^{(\alpha)}$

is the dilation of $T_1^{(\alpha)}$ by $t^{1/\alpha}$, and $T_s^{(\alpha)} \boxplus T_t^{(\alpha)} = T_{s+t}^{(\alpha)}$; so $(T_t^{(\alpha)})_{t\ge 0}$ is a free strictly α-stable semigroup. In the parametrization of [4] it is $\nu_{\alpha,1}$ dilated by $t^{1/\alpha}$; it is concentrated on $(0,\infty)$ [4, Corollary 1] and absolutely continuous [1].

We use the results of Huang and Wang [6, Section 3] for the free convolution $\mu \boxplus \nu$ of a probability measure μ with a nondegenerate freely infinitely divisible ν. Put

$$H(z) = z + \varphi_\nu(F_\mu(z)), \qquad z \in \mathbb{C}^+. \tag{3.1}$$

It has the form $H(z) = a + z + \int (1+sz)/(z-s)\, d\sigma(s)$ with a finite measure $\sigma \neq 0$, and the function $f(x) = \inf\{y > 0 : \int (1+s^2)/((x-s)^2+y^2)\, d\sigma(s) < 1\}$ is continuous and nonnegative. Let $V = \{x \in \mathbb{R} : f(x) > 0\}$. The facts we need are the following.

(HW1) [6, Proposition 3.1(5)] The map $h(x) = H(x + if(x))$ is a strictly increasing homeomorphism of $\mathbb{R}$ onto $\mathbb{R}$.

(HW2) [6, Proposition 3.2] For $x \in \mathbb{R}$: $x \notin V$ if and only if the angular derivative $H'(x)$ exists in $[0,+\infty)$; here $H'(x)$ is defined when H has a real nontangential limit $H(x)$ at x, as the nontangential limit of $(H(z)-H(x))/(z-x)$, and it equals the ordinary derivative when H is analytic and real near x.

(HW3) [6, Theorem 3.6, Remark 3.7] $\mathbb{R} \setminus V = \mathcal{A} \cup \mathcal{B} \cup \mathcal{C}$, where $\mathcal{A}$ is the set of x with $F_\mu(x) = 0$ and $\mu(\{x\}) + \nu(\{s_\nu\}) \ge 1$, s_ν being the zero of F_ν when it exists (when it does not, the term $\nu(\{s_\nu\})$ is read as 0); $\mathcal{B}$ is contained in the set of x at which the vertical limit $G_\mu(x) = \lim_{y\downarrow 0} G_\mu(x+iy)$ exists in $\mathbb{R}\setminus\{0\}$; and $\mathcal{C}$ is contained in the set of x at which this limit is 0.

(HW4) [6, Example 3.8, Theorem 3.9] $\mu \boxplus \nu$ has no singular continuous part, its atoms lie in $h(\mathcal{A})$, and its absolutely continuous part has a density p that is continuous on $\mathbb{R}\setminus h(\mathcal{A})$, with

$$p(h(x)) = \frac{f(x)}{\pi} \int \frac{d\mu(s)}{(x-s)^2 + f(x)^2} \quad (x \in V), \qquad p(h(x)) = 0 \quad (x \in \mathbb{R}\setminus V).$$

3.2. **Proof of Theorem 1.1, free part.** Fix $\alpha \in (0,1)$ and $t > 0$, and write $\mu = \mu_*$, $\nu = T_t^{(\alpha)}$, $G = G_\mu$, $F = F_\mu$. Recall $1 - W \ge 3/4$ and $W \le 1/4$.

Step 1: $\mathcal{A} = \emptyset$, and $\mu \boxplus \nu$ has a continuous density p. Since ν has no atoms, a point of $\mathcal{A}$ would be an atom of μ of mass 1, and μ has none. By (HW4), $\mu \boxplus \nu$ is absolutely continuous with a density p continuous on $\mathbb{R}$.

Step 2: the atoms $-A_k$ lie in V. At $x = -A_k$ we have $|G(x+iy)| \ge |\operatorname{Im} G(x+iy)| \ge w_k/y \to \infty$ as $y \downarrow 0$, so the vertical limit of G at x is not a real number; hence $x \notin \mathcal{B}\cup\mathcal{C}$, and $x \notin \mathcal{A}$ by Step 1. By (HW3), $x \in V$, and by (HW4), $p(h(-A_k)) > 0$.

Step 3: a zero interval in each late gap. Choose $n \ge 2$ with

$$A_n^\alpha \ge 18(1-\alpha)t, \tag{3.2}$$

write $A = A_n$, and put $c = -A/2$. Since $A_{n-1} = \sqrt{A} \le A/4$, we have $-A < c < -A_{n-1}$. We estimate

$$G(c) = \frac{1-W}{c} + \sum_{k\ge 1} \frac{w_k}{c + A_k}, \qquad I_2(c) := \int \frac{d\mu(s)}{(c-s)^2} = -G'(c).$$

For $k \le n-1$, $c + A_k \le -A/2 + A/4 < 0$, so these terms are negative, and $|c + A_k| \ge A/4$. For $k \ge n$, $c + A_k \ge A/2$. Therefore

$$G(c) \le -\frac{2(1-W)}{A} + \frac{2W}{A} \le -\frac{3}{2A} + \frac{1}{2A} = -\frac{1}{A}$$

and

$$I_2(c) \le \frac{4(1-W)}{A^2} + \frac{16W}{A^2} + \frac{4W}{A^2} = \frac{4+16W}{A^2} < \frac{9}{A^2}.$$

Write $a(x) = -G(x)$. The function G is real analytic on $(-A, -A_{n-1})$, and $J_0 = \{x \in (-A, -A_{n-1}) : G(x) < 0\}$ is an open set containing c. For $x \in J_0$, F is analytic near x with $F(x) = -1/a(x) < 0$. The function $w \mapsto w^{1-\alpha}$ on $\mathbb{C}^+$ extends analytically across $(-\infty, 0)$, with boundary value $|w|^{1-\alpha} e^{i\pi(1-\alpha)}$ there. Hence H extends analytically to a neighbourhood of each $x \in J_0$, and on J_0

$$H(x) = x - te^{i\pi\alpha}\, a(x)^{\alpha-1} e^{i\pi(1-\alpha)} = x + t\, a(x)^{\alpha-1} \in \mathbb{R}.$$

Since $a' = -G' = I_2$,

$$H'(x) = 1 - (1-\alpha)\, t\, I_2(x)\, a(x)^{\alpha-2}, \qquad x \in J_0. \tag{3.3}$$

At $x = c$, $a(c) \geq 1/A$ gives $a(c)^{\alpha-2} \leq A^{2-\alpha}$, and with $I_2(c) < 9/A^2$ and (3.2),

$$H'(c) > 1 - 9(1-\alpha)\, t\, A^{-\alpha} \geq \tfrac{1}{2}.$$

By continuity there is an open interval J with $c \in J$ and $\overline{J} \subset J_0$ on which $H' > 0$. At each $x \in J$ the function H is analytic, real on the real axis near x, with real derivative $H'(x) \in (0, \infty)$; so the angular derivative of H at x exists and lies in $[0, \infty)$, and by (HW2), $x \notin V$. By (HW4), $p = 0$ on $h(J)$, which by (HW1) is a nonempty open interval with

$$h(-A_n) < \inf h(J) < \sup h(J) < h(-A_{n-1}).$$

Step 4: no mode. Suppose $\mu \boxplus \nu = c_0\, \delta_a + f_0(x)\, dx$ as in the definition of unimodality. As $\mu \boxplus \nu$ is absolutely continuous, $c_0 = 0$ and $f_0 = p$ almost everywhere. Put $u = h(-A_n)$, $v = h(-A_{n-1})$ and $K = h(J)$. By Steps 2 and 3 and the continuity of p, there are $\delta > 0$ and $\eta > 0$ with $p > \eta$ on $(u - \delta, u + \delta)$ and on $(v - \delta, v + \delta)$, both intervals disjoint from K, with $u + \delta < \inf K$ and $v - \delta > \sup K$. If $a \geq \sup K$, pick $y \in (u - \delta, u + \delta)$ and $z \in K$ with $y < z < a$, $f_0(y) = p(y) > \eta$ and $f_0(z) = p(z) = 0$ (possible since these conditions fail only on a null set); this contradicts $f_0(y) \leq f_0(z)$. If $a < \sup K$, pick $z \in K$ with $z > a$ and $y \in (v - \delta, v + \delta)$, again with $f_0(z) = 0$ and $f_0(y) > \eta$; then $a < z < y$ contradicts $f_0(z) \geq f_0(y)$. Hence $\mu_* \boxplus T_t^{(\alpha)}$ is not unimodal.

The value n in (3.2) depends on α and t, but μ_* does not, so the conclusion holds for every $t > 0$. The check script evaluates G, I_2 and H' at c for $\alpha \in \{0.1, 0.5, 0.9\}$ and $t \in \{1, 100\}$ and on a grid of the gap, and verifies the boundary value of φ used above (checks F1–F4 in [9]). □

Remark 3.1. In the free case the conclusion is stronger than non-unimodality: for every $t > 0$ the support of $\mu_* \boxplus T_t^{(\alpha)}$ has infinitely many gaps, since Step 3 applies to every n satisfying (3.2).

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Declaration of competing interest

The author holds equity in Heights Labs and in Convex Nexus Capital and has a management role at Convex Nexus Capital. These entities are included in the funding declaration above.

Declaration of generative AI and AI-assisted technologies

The paper comes out of Math Brain, an ongoing project on the use of AI systems for research mathematics. On 16 September 2026 agents running in OpenAI's Codex obtained the classical counterexample for the positive 1/2-stable law and the free counterexample for the positive free 1/2-stable law; on 10 October 2026 a further Codex agent restated the classical argument and a Claude agent refereed it. On 17 October 2026 Claude Opus 5.5, working in Claude Code, checked both proofs line by line against the cited theorems, extended them to every $\alpha \in (0, 1)$ with a single initial law, and wrote the computer checks and the paper. The author directed the work and is responsible for the content of the paper and for decisions concerning its dissemination.

Data availability

No external data were used. The code is available at `https://github.com/yspennstate/stable-unimodality-opposite-gaps`.

The paper is part of the Math Brain project, whose public repository is `https://github.com/yspennstate/math-brain-public` [10].

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